

Consequence-Certified State Refinement and Anytime False-Birth Control in POMDP World Models

Aman Agarwal
Sonne Systems
aman@sonnesystems.com

Abstract

Continually learned world models face two coupled errors under partial observability: merging histories whose intervention consequences differ, and creating new latent states or mechanisms whenever prediction error increases. We formulate these errors as controlled decisions. Histories are merged only relative to a declared family of policies and bounded utilities, yielding a query-conditional predictive equivalence related to predictive-state and bisimulation abstractions. Structural growth is admitted only when a predict-before-update alternative accumulates code-weighted sequential likelihood evidence against the deployed model on confirmation observations. For the declared query family, total variation at most ε bounds the expectation error of any utility with range U by $U\varepsilon$. Under an exact calibrated null, predictable alternatives, and a summable candidate code, Ville’s inequality gives anytime control of an erroneous structural birth.

We evaluate narrower components rather than a unified POMDP agent. In a nine-regime linear-Gaussian stream, the confirmation-gated rule recovers the two planted reusable mechanisms in 100% of 200 frozen runs with 0.000 mean false births; a surprise-only policy creates 6.085 mechanisms on average, including 2.085 false births. Recovery is sample limited: at weak effect size, 32 examples recover 0.00 of two mechanisms, whereas 256 recover and compose both in every development seed. On boundary-free supervised Rotated-MNIST, a frozen-base gated adapter obtains 0.9155 and 0.9539 balanced accuracy with 512 and 2,048 adaptation examples, exceeding full replay by 1.98 and 1.23 percentage points across all ten frozen seeds while updating 23.9% as many parameters as the base. A MiniGrid memory diagnostic is deliberately negative: generic recurrent, Transformer, and event-memory models remain at chance, while a typed information-path oracle reaches 100%. These results support confirmation-gated local adaptation and expose identifiability and representation boundaries. They do not establish nonlinear lifetime-valid structural growth, competitive POMDP control, frontier-scale superiority, or an AGI architecture.

1 Introduction

World models used by continual agents must change without confusing three different phenomena: stochastic fu-

tures, hidden-state aliasing, and genuinely new dynamics. A large residual can arise from any of them, as well as from ordinary parameter drift, noise-scale change, observation corruption, or a composition of known mechanisms. A learner that responds to every residual by adding capacity grows without semantic justification. A learner that responds only by increasing invariance may erase distinctions required for control.

This paper studies a narrower question:

When does a partially observed agent have evidence to refine its predictive state or enlarge its mechanism class?

We separate reversible inference from irreversible model-class mutation. Belief updates, routing, uncertainty expansion, retrieval, and temporary local adaptation are reversible. A state split or permanent mechanism birth changes the agent’s ontology and therefore requires a stronger certificate. The state certificate is defined by intervention consequences. The growth certificate is an anytime-valid sequential evidence process with an explicit description code.

This framing is compatible with dense recurrent, Transformer, state-space, or joint-embedding world models. It is not a replacement for them. A dense multi-future predictor remains the default predictive path; structural tests govern when persistent auxiliary structure may be created.

The paper makes five contributions:

1. **Query-conditional predictive equivalence.** We define a resource-bounded state abstraction over intervention-indexed consequence distributions and relate it precisely, but not identically, to goal-conditioned bisimulation.
2. **A consequence certificate.** We state and prove a total-variation bound connecting an allowed merge to error in every declared bounded utility.
3. **An anytime structural-admission certificate.** We show how predictable likelihood-ratio processes and code weights yield false-birth control under optional stopping for a declared candidate family.
4. **An operational decision decomposition.** The controller distinguishes reuse, composition, parameter refinement, uncertainty recalibration, representation repair, abstention, and structural birth.
5. **Falsifiable component evidence.** We report exact frozen results for structural admission, local visual adaptation, partial-observation memory, and dense-versus-routed multi-future prediction.

The non-claims are equally important. The current theory assumes a valid null; it does not solve approximate reuse, adaptively generated lifetime candidate families, or learned conditional calibration. The experiments do not implement one end-to-end nonlinear world model, do not evaluate continuous-control return, and do not compare with frontier language models.

2 Related work

2.1 Predictive states, computational mechanics, and bisimulation

Computational mechanics represents a process by equivalence classes of pasts with the same predictive distribution [1]. Predictive state representations extend this view to controlled systems through predictions of future action-observation tests [2]. Bisimulation and bisimulation metrics preserve reward and transition consequences under state aggregation [3]. DeepMDP learns latent models whose representation errors bound behavioral errors [4], while goal-conditioned bisimulation captures functional equivalence across goals [5]. Value equivalence and action-sufficient representations weaken full predictive equivalence toward distinctions needed by a policy or control objective [6, 7].

Our state definition is closest to an intervention-indexed predictive-state equivalence with a declared utility family. Calling it simply a "goal-conditional bisimulation class" would be imprecise: it operates on histories in a POMDP and compares finite-horizon consequence distributions under policy probes, rather than recursively matching one-step rewards and transitions in an observed MDP. Bisimulation supplies the correct neighboring literature and several possible estimators, not an identity of definitions.

2.2 World models and joint-embedding prediction

DreamerV3 demonstrates a strong dense latent world-model baseline across diverse control domains [8]. V-JEPA 2 shows that large-scale joint-embedding video prediction can support action-conditioned latent planning after limited robot-data post-training [9]. These systems establish that representation, dynamics, and planning can be learned in a dense model at scale. Neither objective by itself decides when a continual learner should create a persistent mechanism or split a state.

Deterministic prediction can average incompatible futures. Multi-head, distributional, and latent-variable models address this by representing alternative futures explicitly. Our local boundary study reinforces a modest design rule: unordered stochastic futures should be represented in the predictive distribution, not forced to coincide with a categorical mechanism router. This is an empirical boundary, not a theorem about all routed models.

2.3 Continual adaptation, changepoints, and dynamic expansion

Bayesian online changepoint detection maintains a posterior over run length [10]. MOCA differentially combines this posterior with meta-learning in unsegmented task streams [11]. CoPE uses probe experts to decide where a continual vision-language model should expand and adds task-location machinery for inference without task labels [12]. COMET and related modular world models learn or compose independent mechanisms [13]. Dynamic expansion, replay, adapters, and mixture-of-experts routing are substantial prior art.

The distinction pursued here is between detecting distribution change and certifying structural novelty. A changepoint can justify investigation without showing that reuse, composition, refinement, recalibration, or representation repair are inadequate. The present experiments compare confirmation-gated growth with surprise-only growth, but not yet with all of MOCA, CoPE, COMET, and modern dynamic-expansion baselines.

2.4 Anytime-valid testing and description length

Likelihood-ratio martingales and Ville's inequality give time-uniform Type-I error control under optional stopping [14]. E-values and e-processes extend this logic to safe sequential testing and optional continuation [15, 16]. Deep anytime-valid tests use predictable learned betting functions for nonparametric problems [17]. Minimum description length (MDL) supplies a principled code cost for model selection [18].

We use these tools rather than claiming them as new. The useful synthesis is to interpret a permanent mechanism birth as a sequentially tested, code-weighted model-class mutation. The theorem below covers exact conditional nulls and a declared candidate code. Approximate learned modules and unrestricted adaptive candidate generation remain open.

3 Formal framework

3.1 POMDP histories and consequence queries

Let a POMDP be

$$\mathcal{P} = (\mathcal{X}, \mathcal{A}, \mathcal{O}, T, \Omega, r, \gamma),$$

with latent state X_t , action A_t , observation O_t , transition kernel T , and observation kernel Ω . The observable history is

$$H_t = (O_0, A_0, \dots, A_{t-1}, O_t).$$

A consequence query is $q = (\pi, g, K)$, where π is a feasible policy probe, g is a bounded utility functional, and K is a finite horizon. It induces the random variable

$$Y_q(h) = g(O_{t+1:t+K}, A_{t:t+K-1}) \mid H_t = h, \text{do}(\pi).$$

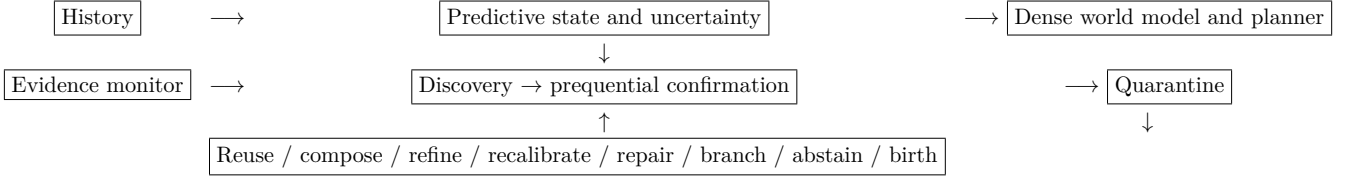


Figure 1: Reversible dense inference is separated from evidence-gated state refinement and structural admission. A quarantined candidate can affect the authoritative library only after a valid confirmation process and protected checks.

Let P_q^h denote its conditional law. For probability measures P, Q on the same measurable space, total variation is

$$\text{TV}(P, Q) = \sup_B |P(B) - Q(B)| = \frac{1}{2} \int |dP - dQ|.$$

For a declared query family $\mathcal{Q}$, define

$$d_{\mathcal{Q}}(h, h') = \sup_{q \in \mathcal{Q}} \text{TV}(P_q^h, P_q^{h'}).$$

Histories are exactly equivalent when this distance is zero and are ε -equivalent when it is at most ε . The definition is deliberately relative to representable goals, safe probes, and an explicit horizon. It makes no promise about queries outside $\mathcal{Q}$.

3.2 Aleatoric branching and epistemic aliasing

Suppose a history supports multiple future modes. We call the ambiguity **operationally aleatoric** relative to $\mathcal{Q}$ when no allowed probe changes the predictive law enough to improve a declared decision at its cost. We call it **probe-resolvable hidden-state aliasing** when a feasible probe changes the posterior over consequence-relevant latent modes and improves a declared decision. These are operational categories, not metaphysical claims: expanding the probe family can turn previously unresolved ambiguity into resolvable aliasing.

A world model should represent unresolved branching as a distribution or set of futures. It should split a predictive state only after a separating consequence is confirmed. Failure to find a separator is not proof that none exists.

3.3 Structural closure before birth

Let $\mathcal{M}_t$ be the authoritative mechanism library. Before adding a mechanism, the controller tests the explanatory closure

$$\mathcal{C}_t = \mathcal{C}_{\text{reuse}} \cup \mathcal{C}_{\text{compose}} \cup \mathcal{C}_{\text{refine}}(\varepsilon) \cup \mathcal{C}_{\text{recalibrate}} \cup \mathcal{C}_{\text{repair}} \cup \mathcal{C}_{\text{branch}}.$$

The possible outcomes are:

- **reuse**: an existing mechanism remains adequate;
- **compose**: compatible existing mechanisms jointly explain the data;
- **refine**: a permitted parameter neighborhood is sufficient;

- **recalibrate**: the mean structure is adequate but uncertainty changed;
- **repair or split**: a probe reveals consequence-relevant hidden state;
- **branch**: multiple futures remain after cost-effective probes;
- **abstain or quarantine**: the model family or sensor contract is inadequate;
- **birth**: a new candidate beats a statistically valid null construction for the declared closure and becomes the quarantined object that earned that evidence.

This decision set prevents distribution change from being used as a synonym for structural novelty.

4 Consequence and structural-admission certificates

4.1 Proposition 1: consequence error under a state merge

Proposition 1. Let h, h' satisfy $d_{\mathcal{Q}}(h, h') \leq \varepsilon$. If every query utility has range length at most U , then, for every $q \in \mathcal{Q}$,

$$|\mathbb{E}_{P_q^h}[Y] - \mathbb{E}_{P_q^{h'}}[Y]| \leq U\varepsilon.$$

Proof. For a fixed q , let the utility lie in $[a, a + U]$ and define $f = (Y - a)/U \in [0, 1]$. The variational characterization of total variation gives

$$|\mathbb{E}_{P_q^h} f - \mathbb{E}_{P_q^{h'}} f| \leq \text{TV}(P_q^h, P_q^{h'}).$$

Multiplying by U , using the definition of $d_{\mathcal{Q}}$, and taking the declared bound proves the result. $\square$

The proposition is elementary but useful: an embedding distance alone does not certify a decision consequence. In practice, estimating the supremum requires a finite query class, calibrated distributional models, or confidence bounds.

4.2 Proposition 2: anytime false-birth control

Let $(\mathcal{F}_t)_{t \geq 0}$ be the filtration before outcome Y_t is revealed. Under the structural null, assume the conditional density is $p_{0,t}(\cdot | \mathcal{F}_{t-1})$. A finite or countable family of candidates is indexed by j . Candidate j supplies a normalized alternative density $p_{1,j,t}$ that is $\mathcal{F}_{t-1}$ -measurable and absolutely continuous with respect to the null density: it may learn from past confirmation outcomes, but must predict before observing Y_t . Define

$$L_{j,n} = \prod_{t=1}^n \frac{p_{1,j,t}(Y_t | \mathcal{F}_{t-1})}{p_{0,t}(Y_t | \mathcal{F}_{t-1})}.$$

Assign nonnegative code weights w_j with $\sum_j w_j \leq 1$; for example, $w_j \propto e^{-K_j}$, where K_j is an MDL-style candidate code.

Proposition 2. Under the stated null and predictability assumptions, $E_n = \sum_j w_j L_{j,n}$ is a nonnegative martingale or supermartingale with $E_0 \leq 1$, and

$$\Pr_0 \left(\sup_{n \geq 0} E_n \geq \frac{1}{\alpha} \right) \leq \alpha.$$

Consequently, admitting candidate j only when

$$\log L_{j,n} \geq \log(1/\alpha) - \log w_j$$

controls the probability of at least one erroneous birth in the declared family by α .

Proof. Conditional on $\mathcal{F}_{t-1}$, normalization gives

$$\mathbb{E}_0 \left[\frac{p_{1,j,t}(Y_t | \mathcal{F}_{t-1})}{p_{0,t}(Y_t | \mathcal{F}_{t-1})} \middle| \mathcal{F}_{t-1} \right] = \int p_{1,j,t}(y | \mathcal{F}_{t-1}) dy = 1.$$

Thus each $L_{j,n}$ is a nonnegative martingale. A nonnegative weighted sum with initial value at most one is a nonnegative supermartingale. Ville’s inequality yields the crossing bound. If any $w_j L_{j,n} \geq 1/\alpha$, then $E_n \geq 1/\alpha$, proving the family-wise statement. $\square$

This proposition permits optional stopping, not arbitrary data reuse. It fails when the null is misspecified, the current outcome tunes its own alternative, or candidates are added without predictable wealth allocation. An MDL penalty is statistically meaningful here only when implemented as a valid code or prior weight; subtracting a post-hoc complexity term from an invalid likelihood ratio does not restore validity.

4.3 Reversible inference versus irreversible mutation

The deployed controller has two timescales. The reversible path updates beliefs, retrieves episodes, expands predictive uncertainty, routes through existing modules, and may activate a quarantined adapter. The irreversible path changes the authoritative state partition or mechanism library. It requires disjoint or prequential confirmation, a registered evidence budget, exact serialization of the crossing candidate, protected regression checks, and rollback metadata.

The learner that earns evidence must be linked to the admitted snapshot. Object identity by itself is insufficient if later updates change the candidate. A practical invariant is

$$\begin{aligned} \text{witness at threshold} &= \text{serialized crossing state} \\ &= \text{admission candidate.} \end{aligned}$$

4.4 Operational algorithm

For each suspected regime change:

1. Keep the dense world model active and raise calibrated uncertainty.
2. On discovery data, retrieve candidate reuse/composition/refinement models and propose probes that distinguish branching from aliasing.
3. Freeze the candidate family or allocate predictable wealth to online candidates.
4. On confirmation observations, predict before update and accumulate the code-weighted evidence process.
5. Reuse, compose, refine, recalibrate, repair, branch, or abstain whenever one of those explanations is adequate.
6. If a new candidate crosses its evidence boundary and protected checks pass, serialize the exact crossing state into quarantine.
7. Consolidate only after a separate deployment evaluation; failed candidates never enter the authoritative library.

The present experiments implement subsets of this algorithm. They do not yet provide the complete nonlinear POMDP system described here.

5 Structural-growth diagnostic

5.1 Protocol

The controlled system is a four-output linear Gaussian process with six state variables and a binary action. Two planted residual mechanisms modify orthogonal action- and state-conditioned directions. Nine unlabeled regimes contain base dynamics, harmless covariate shift, a five-fold noise-standard-deviation shift, mechanism A, a base return, A under shifted state support, mechanism B, A+B, and unpredictable corruption.

Each regime provides 128 discovery and 128 confirmation transitions. A ridge candidate has 32 parameters. The confirmation threshold combines $\alpha = 0.01$ with a half-BIC code. Predictive variance is estimated from discovery residuals, never confirmation outcomes. The protocol was frozen and evaluated on seeds 1800–1999. A surprise-only baseline creates one adapter whenever the old calibrated residual threshold fires.

5.2 Frozen result

For mechanism A, mean squared error decreases from 0.1505 under the base to 0.0106 under the admitted model. For B it decreases from 0.1502 to 0.0107. Noise and corruption do not create persistent modules. The comparison isolates one point: surprise detects change, whereas confirmation against a reuse-aware null can reject changes that do not earn reusable predictive structure.

5.3 Signal and sample boundary

Across 100 development seeds per row, the method can miss the planted factors or fit a visible composite at low sample size. The sharp transition is an identifiability boundary, not evidence of a universal sample threshold. The generator is linear, Gaussian, additive, and aligned

Table 1: Structural decisions across 200 frozen seeds.

Quantity	Confirmation-gated growth	Surprise-only growth
Mean library births	2.000	6.085
Mean false births	0.000	2.085
Exact two-mechanism library	100.0%	not applicable
A revisit reused without growth	100.0%	no reuse test
A+B composed without growth	100.0%	no composition test

Table 2: Development-set signal and sample boundary.

Examples per split	Mean true births of 2	Mean false births	Reuse	Composition
32	0.00	0.00	0%	0%
64	0.25	0.77	7%	3%
128	1.95	0.05	100%	95%
256	2.00	0.00	100%	100%
512	2.00	0.00	100%	100%

with the candidate class. The experiment does not validate approximate module nulls, neural likelihood calibration, or lifetime adaptive testing.

6 Boundary-free Rotated-MNIST adaptation

6.1 Protocol

A 105,866-parameter convolutional classifier is pre-trained on unrotated MNIST. The adaptation stream contains labeled 45-degree images without a task, domain, or boundary identifier. An old-data-only energy detector averages 32 stochastic augmentations and activates a 25,274-parameter residual adapter; the base is frozen. Evaluation uses all 10,000 official test images in both domains.

The frozen protocol uses ten seeds (100–109), 20,000 base-training examples, 4,000 calibration examples, and 12 repeated adaptation passes. It is therefore boundary-free supervised adaptation, not strict single-pass continual learning. Baselines include always-on adaptation, full fine-tuning, last-layer tuning, EWC, Learning without Forgetting, and replay variants. Full replay stores one old example per new example and receives twice as many optimizer examples as the gated adapter.

6.2 Frozen result

The 95% paired bootstrap intervals are [1.57, 2.38] percentage points at 512 examples and [0.97, 1.48] at 2,048. The detector activates on 2.72% of old samples and 100.0% of rotated samples. The adapter updates 23.9% of the base parameter count and stores no old examples.

This is the strongest recognized positive result in the paper, but its scope is narrow. The shift is visually simple, labeled, and almost perfectly separable; the experiment has only one new domain and no mechanism library, planning, or active intervention. It supports frozen-base gated adaptation, not general structural self-modification.

7 Representation boundary studies

7.1 MiniGrid hidden-state aliasing

MiniGrid-MemoryS17Random presents a cue, a hallway, and a T-junction. The evaluation harness first views the cue and later reaches a junction whose current observation omits it. The supervised label is the optimal upper or lower branch. Training and test seeds are disjoint and class balanced.

The typed model receives MiniGrid’s object ontology, the rule that the cue is a single visible object, and an exact type-matching operator; it learns only a temperature and two biases. It is an information-path oracle, not a fair neural baseline. The result shows that the trajectory contains sufficient information when the relevant variable and comparison are supplied. It simultaneously falsifies the claim that the tested generic history encoders discover that variable from these sample budgets. This is an empirical architectural boundary, not a theorem that recurrence or attention cannot solve the task.

7.2 Dense versus routed multi-future prediction

In a separate parameter-matched synthetic physical-latent set-prediction study, the stable 21,608-parameter dense multi-future model obtains 9.400×10^{-6} mean out-of-distribution set MSE across five fresh seeds. A 16,476-parameter posterior two-axis routed model obtains 1.152×10^{-5} ; a 16,892-parameter smaller dense match obtains 1.476×10^{-5} but is unstable across seeds. The posterior router’s out-of-distribution exact route-set accuracy is 0.0167%.

The full dense model beats the posterior routed model on every matched fresh seed. The smaller dense model’s worse mean is driven by instability, so it does not justify claiming a routing advantage. The defensible conclusion is local: on this unordered multi-future target, predictive accuracy did not require semantic route recovery, and the tested discrete routers did not provide a stable advantage. Dense distributional prediction should remain the default until routing earns a measured benefit.

Table 3: Boundary-free Rotated-MNIST adaptation against full replay.

New-domain examples	Gated adapter	Full replay	Paired gain	Wins	Exact sign-flip p
512	0.9155	0.8957	1.98 pp	10/10	0.00195
2,048	0.9539	0.9415	1.23 pp	10/10	0.00195

Table 4: MiniGrid hidden-state diagnostic on disjoint, balanced test seeds.

Training episodes	Final frame	GRU	Transformer	Event memory	Typed oracle
64	50.0%	50.0%	50.0%	49.5%	100.0%
256	50.5%	49.1%	49.1%	49.0%	100.0%
1,024	50.0%	50.3%	49.8%	50.2%	100.0%

7.3 What the negative results establish

Negative outcomes constrain the design without becoming "boundary theorems." The MiniGrid result says that a supplied sufficient statistic can dominate generic learning at low sample count, not that hand-designed structure is universally superior. The multi-future result says that the tested routers lose to a strong dense model, not that every modular architecture must collapse. Structural bias should therefore be evaluated by the consequence it certifies and the generalization it improves, rather than by architectural elegance.

8 Limitations, continuous-control path, and conclusion

8.1 Theoretical limitations

The total-variation certificate is only as broad as $\mathcal{Q}$. High-dimensional TV is hard to estimate, and an untested query can reveal a harmful merge. The sequential theorem assumes an exact conditional null. A learned module that is merely close to the truth will eventually be rejected unless the null includes an explicit refinement neighborhood. Code-weighted candidate families do not by themselves solve unrestricted adaptive library growth, candidate-selection leakage, dependent heavy-tailed errors, or online false discovery over an unbounded lifetime.

The evidence-generating predictor and stored mechanism must also remain linked at the crossing time. The linear diagnostic uses a narrow discovery/confirmation construction; the full evidence-witness invariant, approximate-null theory, and lifetime wealth accounting are research requirements rather than completed contributions.

8.2 Empirical limitations

The structural benchmark is synthetic and aligned with linear additive mechanisms. Rotated-MNIST is supervised and nearly separable. The MiniGrid experiment is trajectory classification, not online control, and the tested neural models remain at chance. The multi-future benchmark is synthetic. No experiment uses a learned nonlinear visual world model to choose end to end among reuse, composition, refinement, recalibration, probing, splitting, abstention, and birth.

No energy, latency, memory-traffic, analog-drift, or neuromorphic measurement supports a hardware claim. No result compares capability with a frontier language or multimodal model at matched data and compute.

8.3 Path to continuous-control evaluation

A decisive successor should use MuJoCo or Brax tasks with hidden physical parameters, returning regimes, composable changes, observation corruption, and probe actions. The agent should encounter, without labels:

- exact recurrence of an old mechanism;
- composition of known mechanisms;
- small mass, friction, or actuator refinement;
- observation-noise recalibration;
- multi-future stochasticity;
- probe-resolvable hidden-state aliasing;
- a genuinely withheld transition mechanism.

All methods should share the same encoder, world-model capacity, optimizer budget, observations, and action cost. Required baselines include a fixed dense world model, full fine-tuning, replay, BOC PD, MOCA, dynamic expansion, CoPE-like probe-guided expansion, and an oracle structural controller. Primary endpoints are false and missed births, refinement-as-birth errors, duplicate modules, intervention prediction, control return net of probe cost, library size, runtime, and memory. A fresh registered split and independent reproduction are required before any broad claim.

8.4 Conclusion

The central proposal is a governance rule for learned world models: predictive states should be refined by verified intervention consequences, and persistent structure should be created only by an anytime-valid evidence process against a declared reusable null. Constructing such a null for an approximate learned closure remains open. The consequence and e-process results make the current assumptions explicit. The component experiments show that confirmation can reject surprise-driven births and that a frozen-base local gate can outperform replay on a simple shift. The negative studies show that generic history models can fail to discover a sufficient variable and that routed semantics need not beat dense multi-future prediction.

The evidence is a research foundation, not a completed general agent. The next scientific step is not a stronger

claim; it is an end-to-end learned nonlinear POMDP system with approximate-null and lifetime-valid theory, strong matched baselines, continuous-control consequences, and independent reproduction.

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